# Akshat G

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## Summary

AI/ML engineer and MedTech startup co-founder specializing in production ML systems, computer vision, LLMs, and GPU-accelerated inference. Experience building real-time perception pipelines, leading technical teams, and shipping end-to-end AI systems from research to deployment.

## Education

**Manipal Institute of Technology** — Bengaluru, India
B.Tech (Honours), Computer Science and Engineering — AI Specialization — Jul 2023 – May 2027

- **CGPA:** 9.56/10

## Professional Experience

**Shell India Markets Private Limited** — Bengaluru, India
Research Intern — Multiphysics AI & Product Innovation — Jun 2026 – Present

- Developed HiME (Hierarchical Multi-label Estimator), a novel architecture for hierarchical classification across 27 classes and roughly 1.05M valid label paths, achieving 100% hierarchy consistency and 91% root-to-leaf accuracy on a custom gear defect dataset.
- Architected a multi-agent ensemble inference pipeline simulating an expert panel, coordinating specialized vision backbones (ConvNeXt, DeepLabV3, ResNet-50) to corroborate and contextualize HiME's primary diagnostics.
- Deployed a client-side AI assistant that parses ensemble JSON outputs into a fine-tuned instruct LLM for automated report generation, cutting turnaround from roughly 40 hours to 2 hours with human-in-the-loop validation.

## Featured Projects

**Sensera** – **[Code] [DOI]** — Bengaluru, India
Co-Founder & ML Engineer — Feb 2026 – Jun 2026

- Co-founded an AI MedTech startup; shipped a full-stack Personal Health Record platform with 50+ daily active users.
- Built an async document pipeline decoupling OCR/NLP inference across 10 parallel GPU workers, achieving sub-3-second document processing and extraction.
- Implemented semantic search over medical records (pgvector, row-level security) and a self-hosted LLM pipeline for summarization and structured extraction — sub-second, user-isolated retrieval with no dependence on external inference APIs.

**CEAM — Center of Excellence in Autonomous Mobility** – **[Code] [DOI]** — Bengaluru, India
Perception Lead, Project MAVERIC — Aug 2025 – Aug 2026

- Led perception for MAVERIC, a fully autonomous go-kart, building the real-time perception stack around ROS 2, sensor integration, vision pipelines, and object tracking.
- Led a 16-member computer vision team; built perception pipelines in optimized C++ (Eigen) and NumPy. Achieved 56 FPS semantic segmentation and 16 FPS object detection.
- Developed a custom LiDAR-camera fusion pipeline for 3D scene understanding and real-time path planning. mera fusion pipeline for 3D scene understanding and real-time path planning.

## Publications [Google Scholar] [ORCID]

2 conference and 5 accepted journal papers spanning data-centric AI, semi-supervised learning, representation learning, efficient ai, and explainable ai including Scientific Reports, SAGE Digital Health, Springer CCIS, and IEEE Access.

## Technical Skills

- **Programming:** Python, C/C++, Java, TypeScript/JavaScript, SQL, Bash
- **ML Frameworks & Tools:** PyTorch, TensorFlow, HuggingFace Transformers, Sentence-Transformers, TensorRT, ONNX Runtime, CUDA, OpenCV, scikit-learn, NumPy
- **AI/ML:** Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, RAG, Semantic Search, Vector Embeddings, Transfer Learning, Fine-tuning
- **Cloud & Systems:** Linux, Docker, Modal, AWS, Google Cloud, Supabase, Git
- **Databases & Backend:** PostgreSQL (Schema Design, RLS, Migrations, Indexing), pgvector (HNSW ANN), FastAPI, Pydantic, React, React Native